

$$\delta = \theta_{i1} + \sin^{-1} \left[\sin \alpha (n^2 - \sin^2 \theta_{i1})^{\frac{1}{2}} - \sin \theta_{i1} \cos \alpha \right] - \alpha$$

The deviation suffered by a monochromatic beam on traversing a given prism is a function only of the incident angle at the first face.

The smallest value of δ is known as the **minimum deviation**

Starting from

$$\delta = \theta_{i1} + \theta_{t2} - \alpha$$

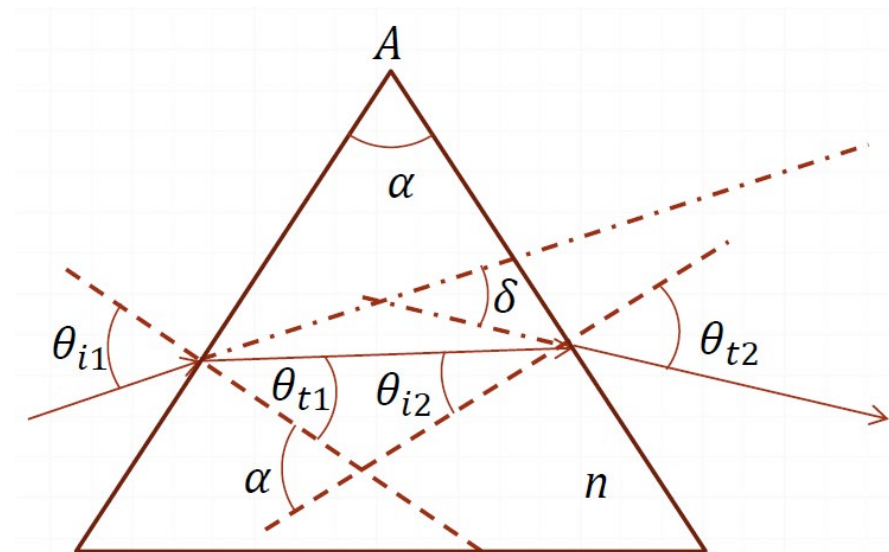
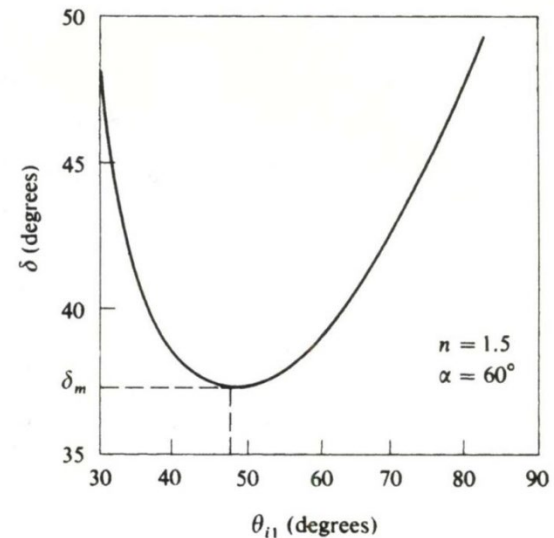
$$\alpha = \theta_{t1} + \theta_{i2}$$

Solution:

$$\theta_{i1} = \theta_{t2}$$

$$\theta_{t1} = \theta_{i2}$$

The ray for which the deviation is a minimum traverses the prism symmetrically, that is parallel to its base.



In the case when $\delta = \delta_m$

$$\delta_m = \theta_{i1} + \theta_{t2} - \alpha$$

$$\theta_{i1} = \theta_{t2}$$

$$\theta_{i1} = (\delta_m + \alpha)/2$$

$$\alpha = \theta_{t1} + \theta_{i2}$$

$$\theta_{t1} = \theta_{i2}$$

$$\theta_{t1} = \alpha/2$$

Snell's law

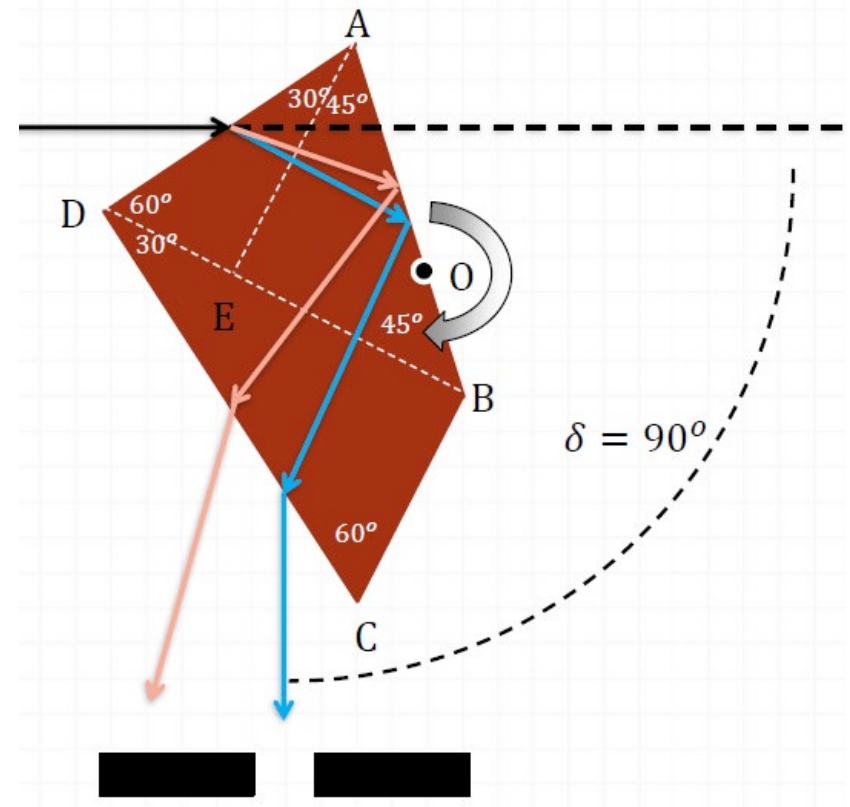
$$n = \frac{\sin \theta_{i1}}{\sin \theta_{t1}} = \frac{\sin (\delta_m + \alpha)/2}{\sin \alpha/2}$$

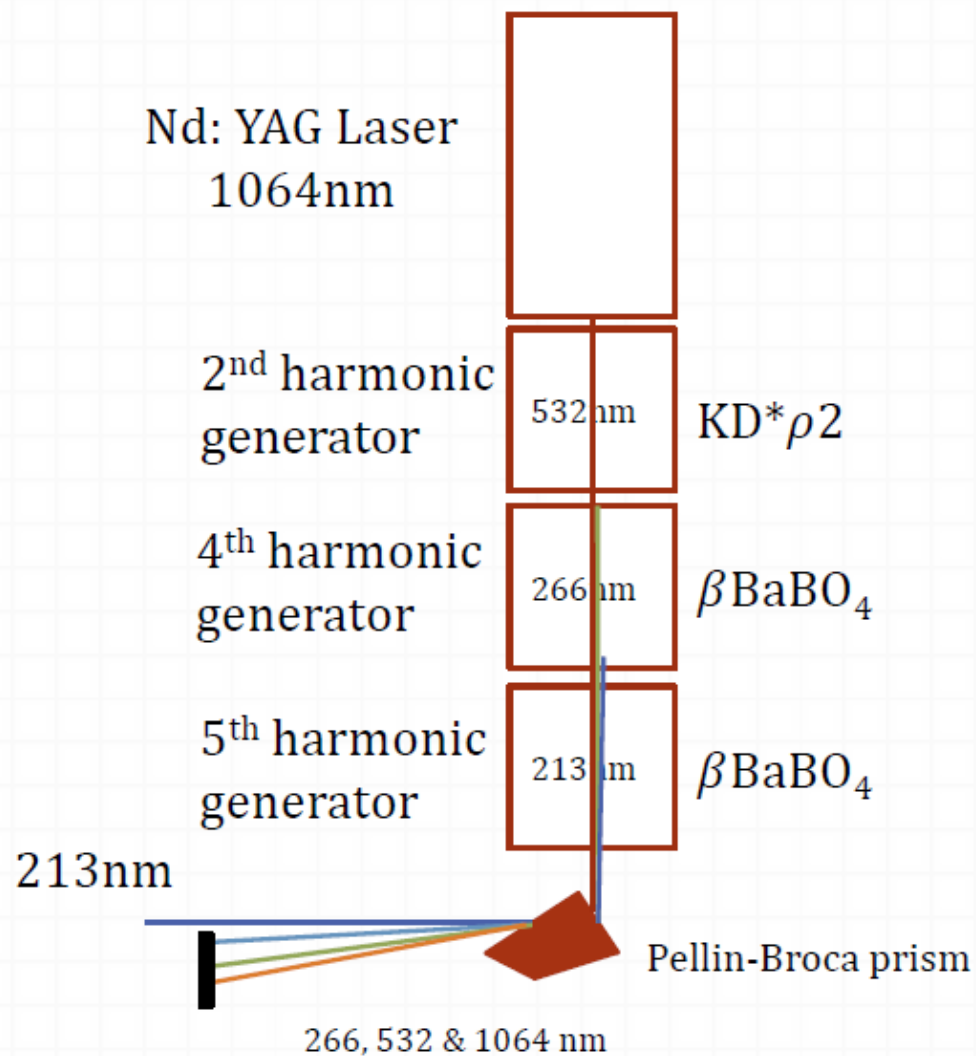
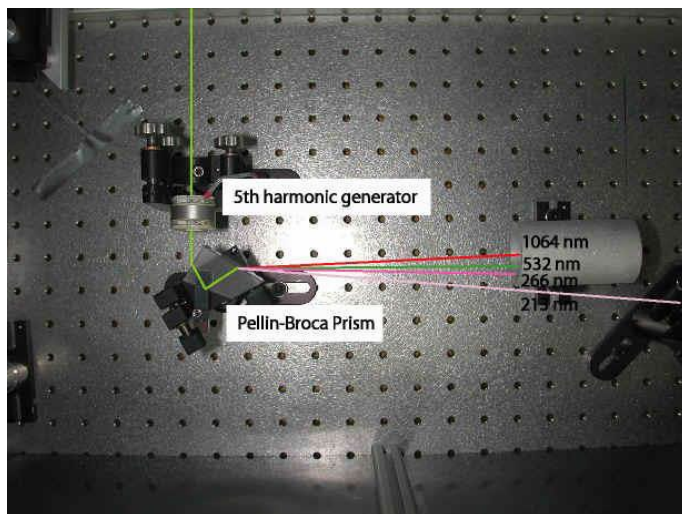
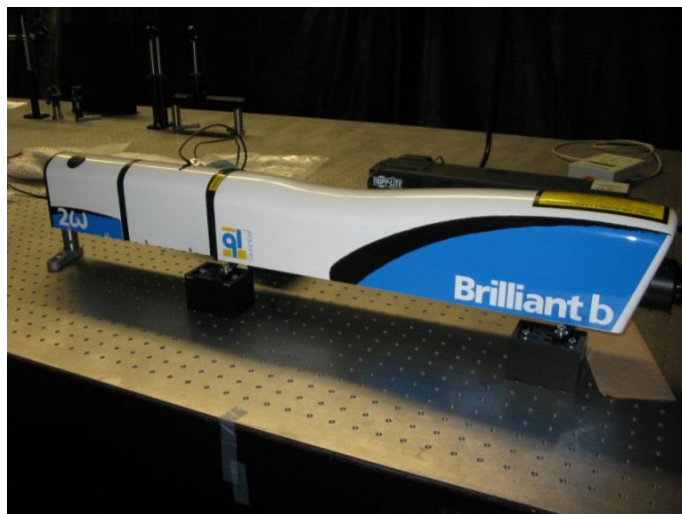
This equation forms the basis of technique for determining the refractive index.

Constant-deviation dispersing prisms (恒偏向色散棱镜)

Pellin-Broca (佩林-布洛卡) prism

- Suppose that in the position shown a single monochromatic ray of wavelength λ traverses the component prism DAE symmetrically, thereafter to be reflected at 45° from face AB. The ray will then traverse prism CDB symmetrically, having experienced a total deviation of 90° .
- As the prism is rotated around a point O, the selected wavelength which is deviated by 90° is changed without changing the geometry of relative positions of the input and output beams.





Reflecting prisms

Reflecting prisms: dispersion is not desirable, which means that deviation is independent of λ .

The angle of deviation (δ):

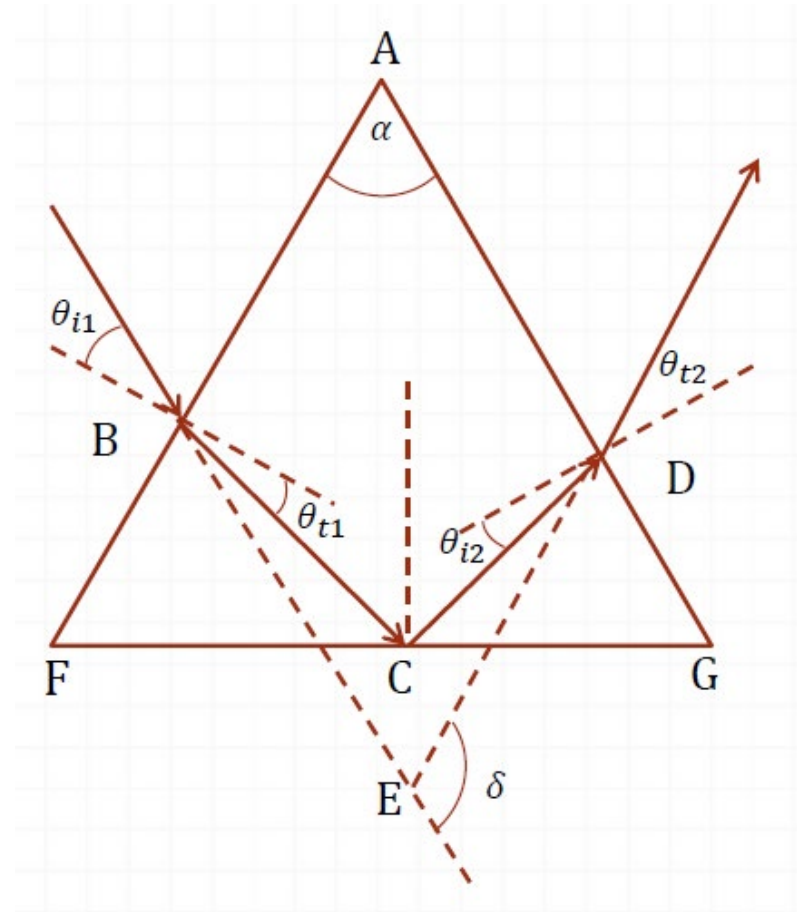
$$\delta = \theta_{i1} + \theta_{t2} + \alpha$$

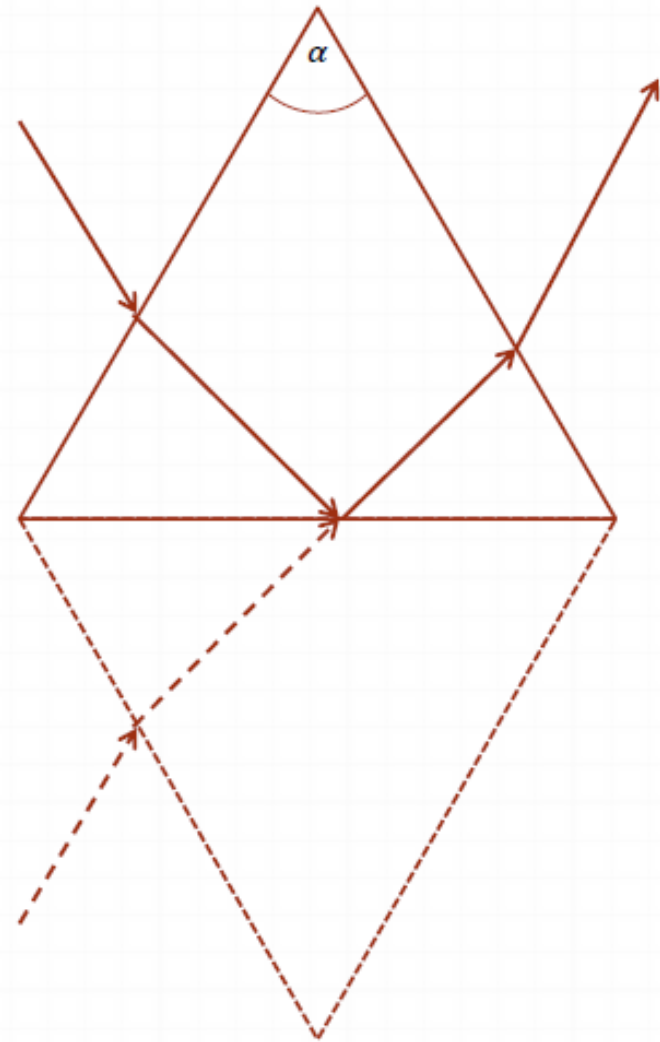
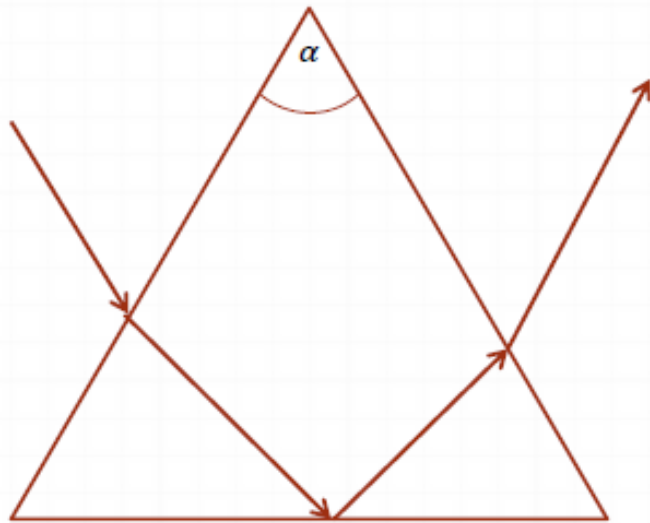
$$\delta = 2\theta_{i1} + \alpha$$

which is independent of both λ and n .

That is to say, it is **achromatic**.

Purpose: change the direction of propagation or the orientation of the image, or both.





The image of the incident ray emerges parallel to itself, regardless of wavelength.

Reflecting prisms

Image Inversion and Reversion by Right-Angle Prisms

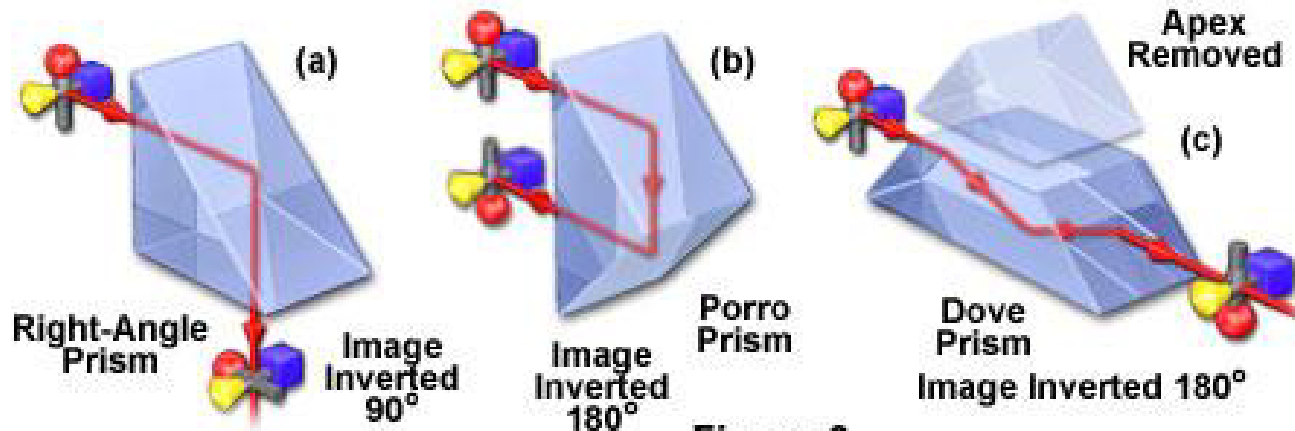


Figure 2

Image Rotation, Deviation, and Displacement by Common Prisms

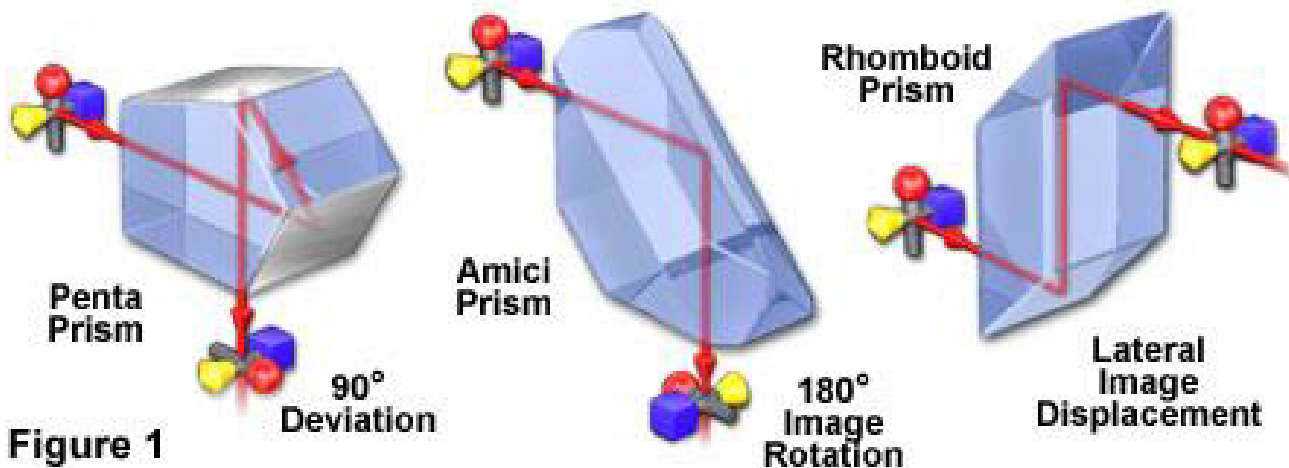


Figure 1

4.6 Fiber optics

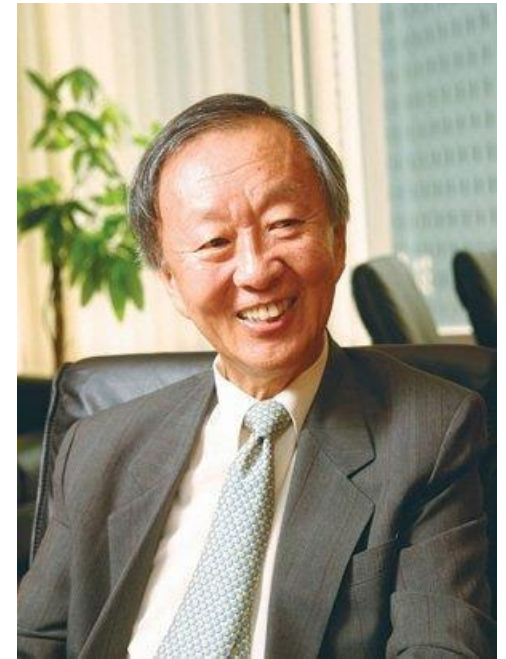
Communication method:

Electric current, microwave, **light**.

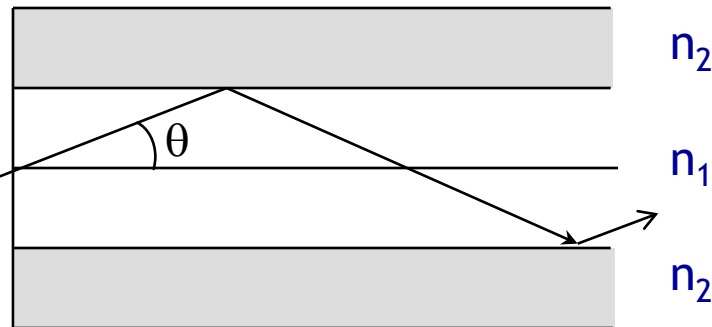
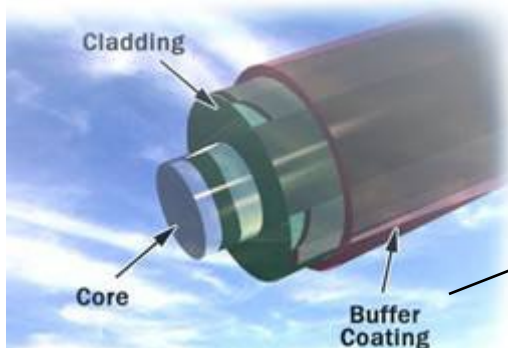
Fiber: because of its low-loss transmission, high-information-carrying capacity, small size and weight, immunity to electromagnetic interference, unparalleled signal security, and the abundant availability of the required raw materials (i.e., ordinary sand), ultra pure glass fibers have become the premier communications medium.

1966 Breakthrough of fiber optics

Charles K. Kao, working with George Hockham at Standard Telecommunication Laboratories in Harlow, England, makes a discovery that leads to a breakthrough in fiber optics. He calculates how to transmit light over long distances via optical glass fibers, deciding that, with a fiber of purest glass, it would be possible to transmit light signals over a distance of 100 km, compared with only 20 m for the fibers available in the 1960s. Kao receives a 2009 Nobel Prize in physics for his work.



1933—2018



The path length:

$$\ell = L / \cos \theta_t$$

From Snell's law:

$$\ell = n_f L (n_f^2 - \sin^2 \theta_i)^{-1/2}$$

The number of reflections N:

$$N = \frac{\ell}{D / \sin \theta_t} \pm 1 = \frac{L \sin \theta_i}{D (n_f^2 - \sin^2 \theta_i)^{1/2}} \pm 1$$

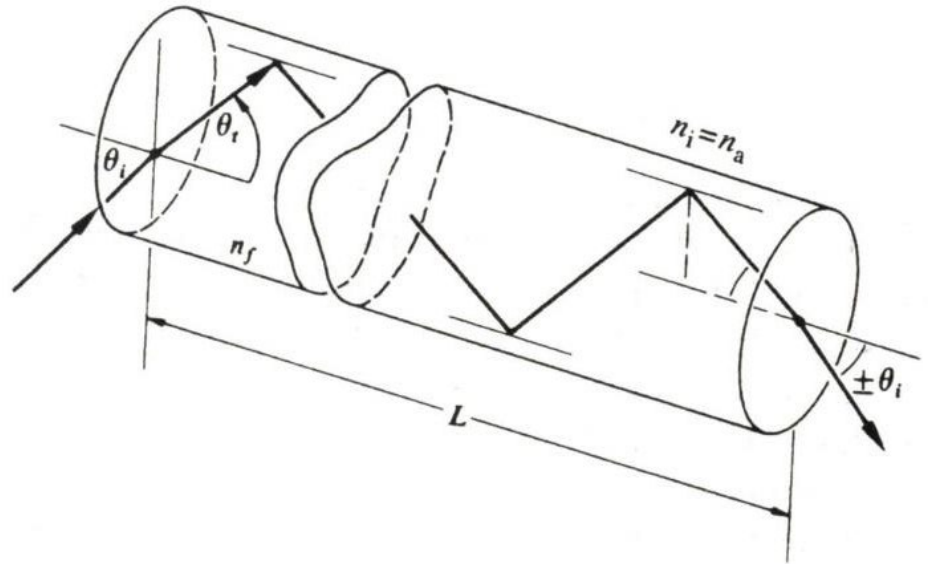
Example: $D = 50 \mu\text{m}$; $n_f = 1.6$; $\theta_i = 30^\circ$
 $N/L = 2000$ reflection per foot.

Leakage of light:

If the surface of a single fiber is covered by moisture, dust, oil (via FTIR)

If large number of fibers are packed in close proximity (via FTIR), i.e., cross-talk

Solution: Each fiber is covered with a transparent sheath of lower index.



Numerical aperture

A fiber core has an index n_f , and the cladding has an index n_c .

$$\sin \theta_c = \frac{n_c}{n_f} = \sin(90^\circ - \theta_t)$$

Thus

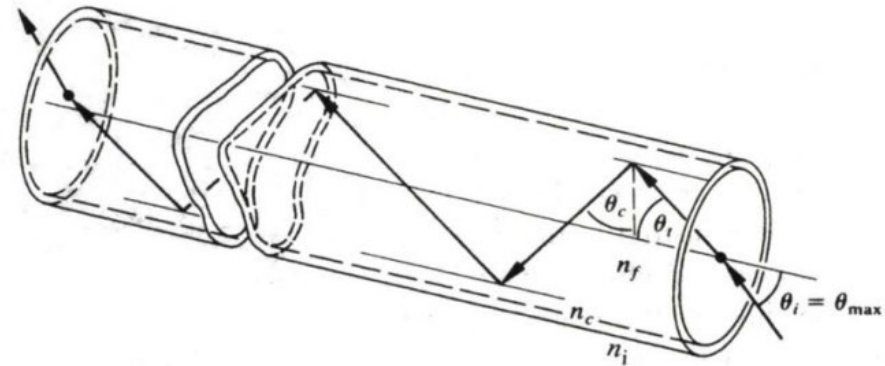
$$\frac{n_c}{n_f} = \cos \theta_t = (1 - \sin^2 \theta_t)^{1/2}$$

Solution:

$$\sin \theta_{max} = \frac{1}{n_i} (n_f^2 - n_c^2)^{1/2}$$

Numerical aperture or NA: $n_i \sin \theta_{max}$

Fibers with a wide variety of numerical apertures, from about 0.2 up to and including 1, are commercially obtainable.



Bundles of free fibers whose ends are bound together, ground, and polished from flexible light guides. If no attempt is made to align the fibers in an ordered array, they form an ***incoherent bundle***.

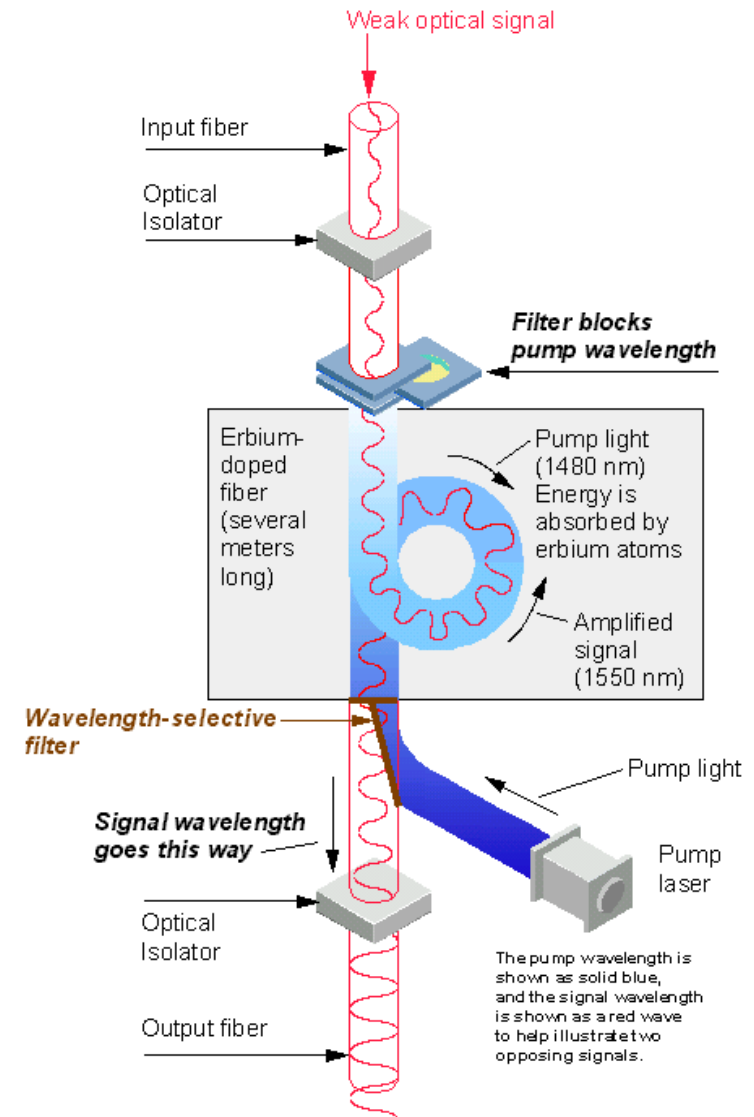
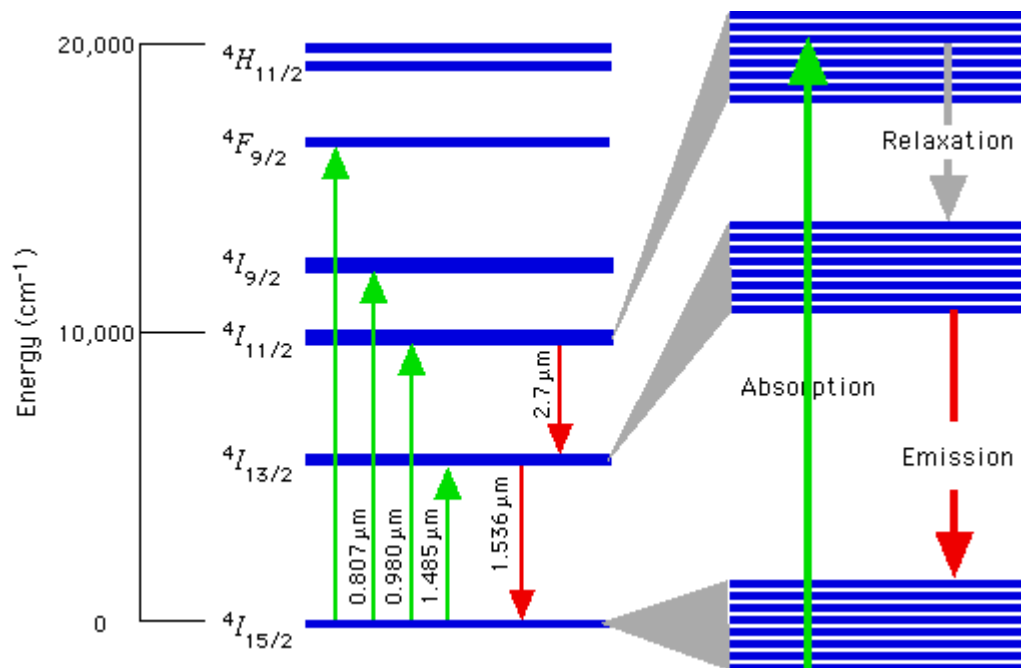


When the fibers are carefully arranged so that their terminations occupy the same relative positions in both of the bound ends of the bundle, it is said to be ***coherent***.



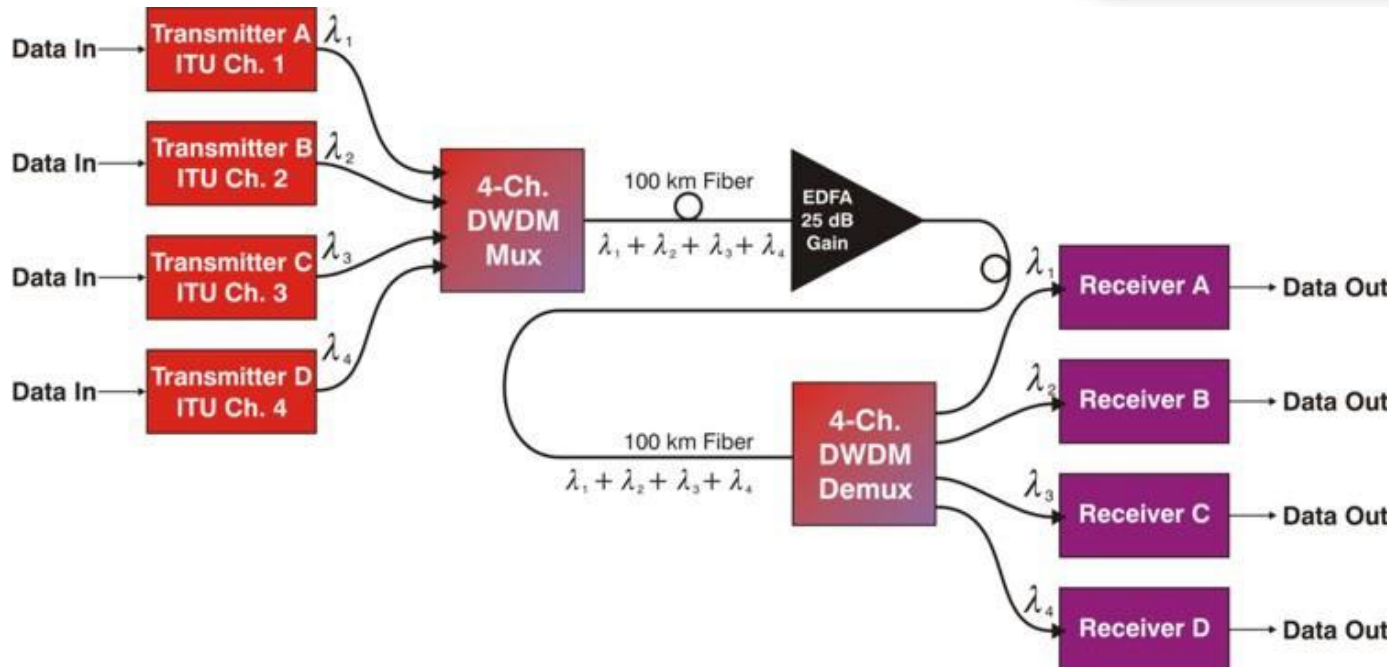
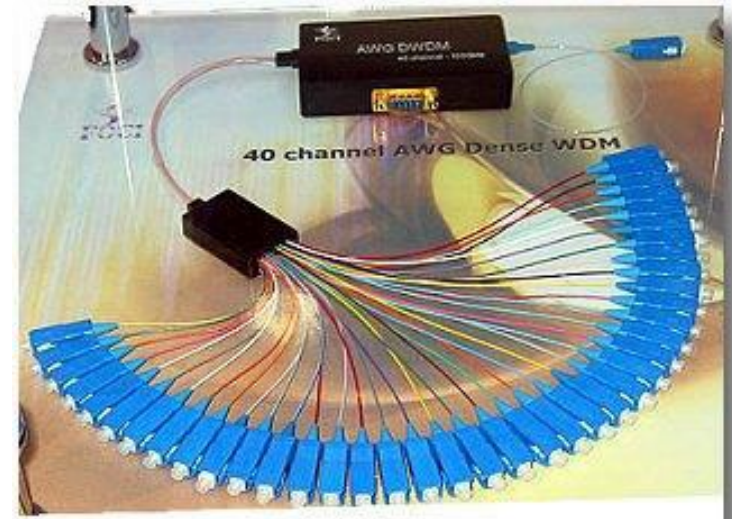
Fiberoptic Communications Technology

- ❑ **Attenuation:** due to impurities (especially the ions of iron, nickel and copper) and contamination of the fibers.
- ❑ **Innovational solution: erbium-doped fiber amplifiers**

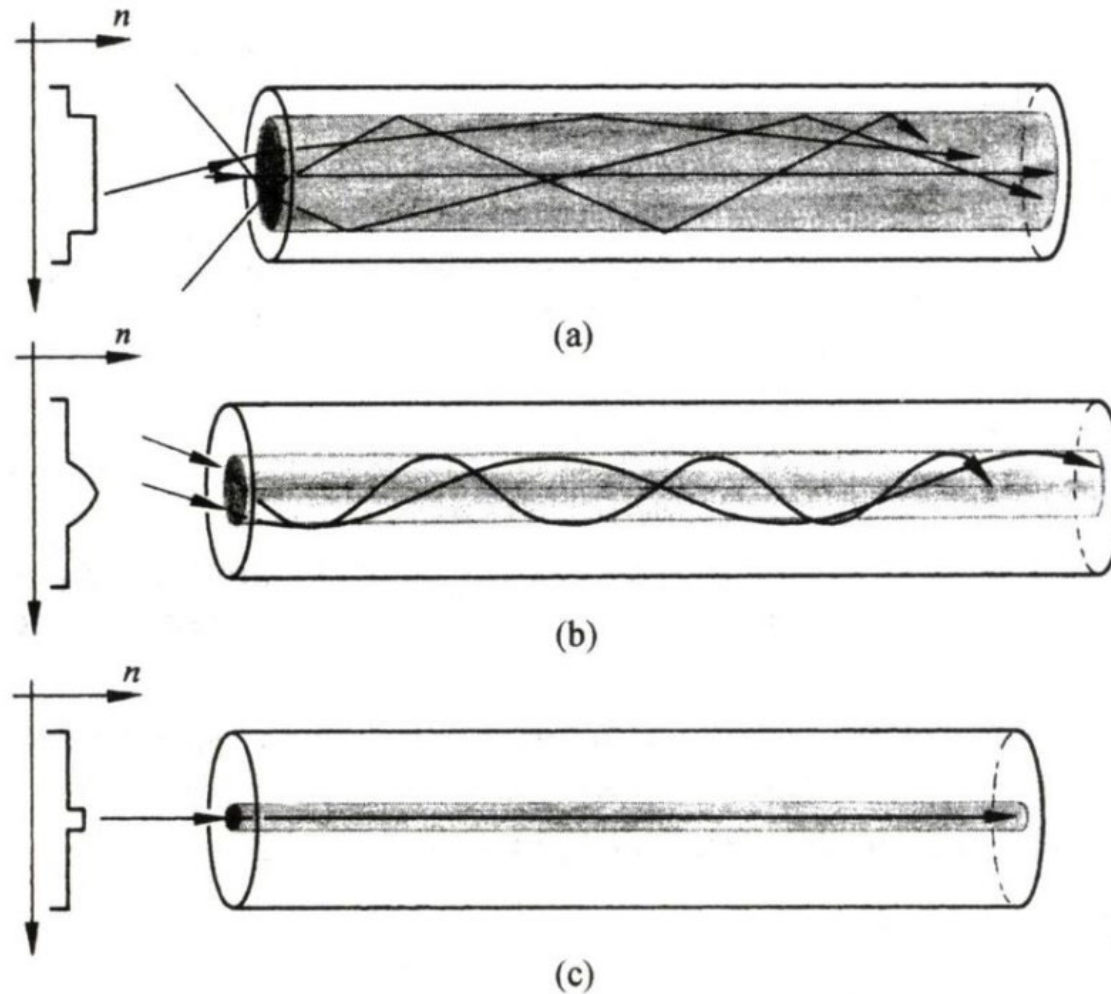


The second innovational technique:

Dense wavelength division multiplexing: all the optical signals transmitted at the same time over the same fiber at different frequencies.



Three major fiber configurations

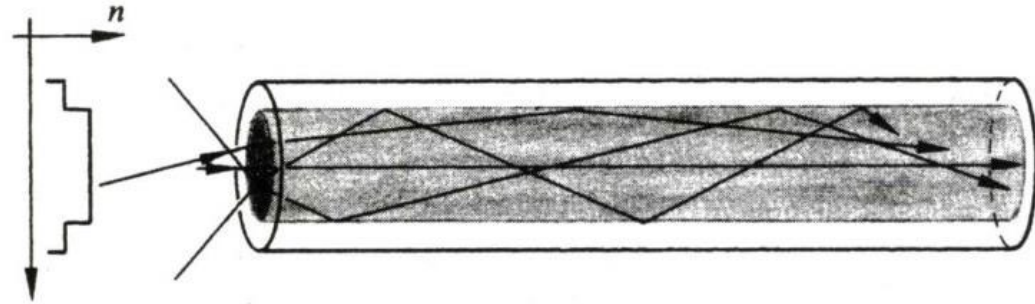


- (a) Multimode step-index fiber.
- (b) Multimode graded-index fiber.
- (c) Single-mode step-index fiber.

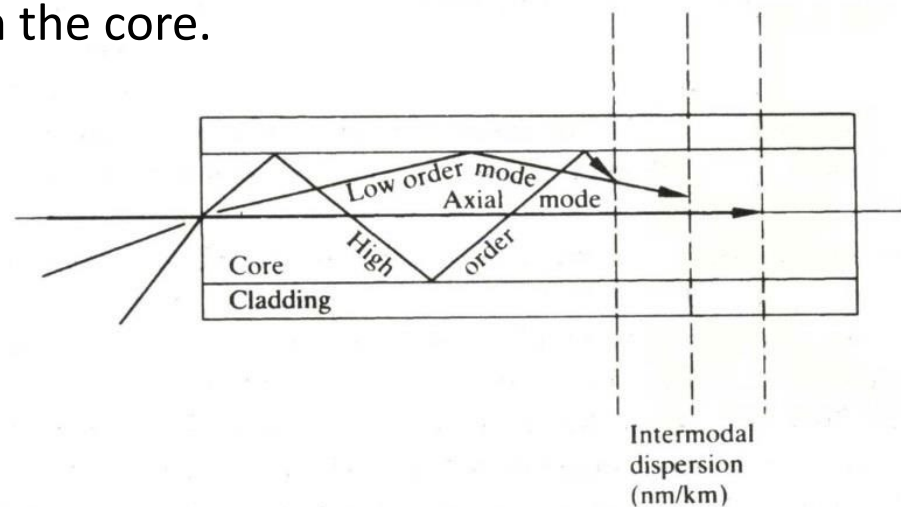
Stepped-index fiber:

A homogeneous core of roughly 50 to 200 μm and cladding typically 20 μm thick.

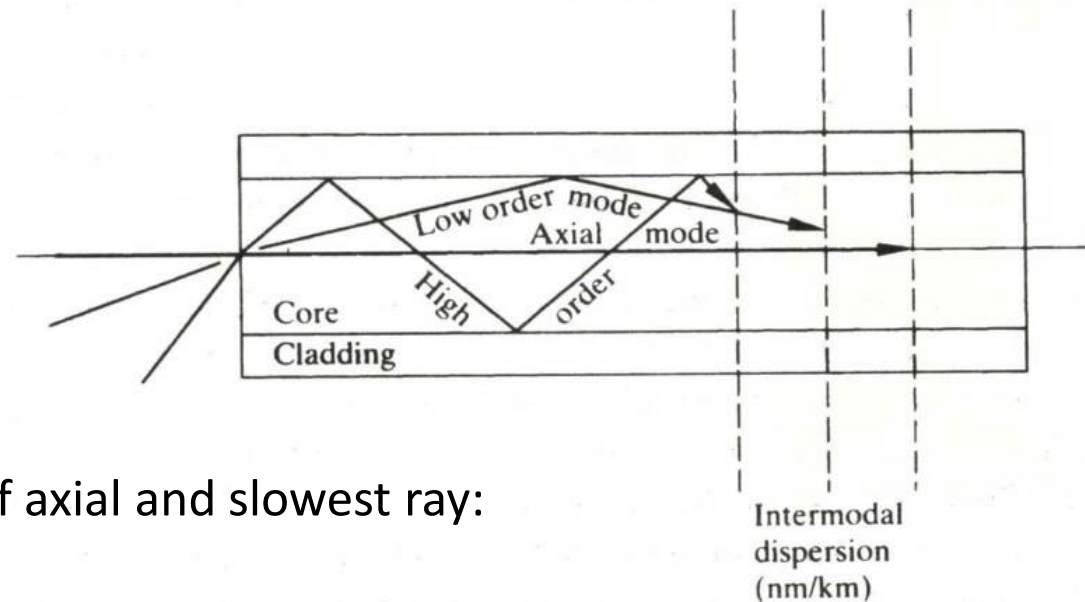
The stepped-index fiber was widely used in the first-generation systems (1975-1980)



Multimode fiber: depending on the launch angle into the fiber, there can be hundreds, even thousands, of different ray paths or modes by which energy can propagate down the core.



Intermodal dispersion: each mode in multimode fiber corresponds to a slightly different transit time.



Time delay between the arrive of axial and slowest ray:

Lowest order mode
$$t_{min} = \frac{L}{v_f} = \frac{L}{c/n_f} = \frac{Ln_f}{c}$$

Highest order mode
$$t_{max} = \frac{\ell}{v_f} = \frac{L/\cos \theta_t}{c/n_f} = \frac{Ln_f/n_c}{c/n_f} = \frac{Ln_f^2}{cn_c}$$

$$\Delta t = \frac{Ln_f}{c} \left(\frac{n_f}{n_c} - 1 \right)$$

Example

The index of core and cladding $n_f=1.5$ and $n_c=1.489$

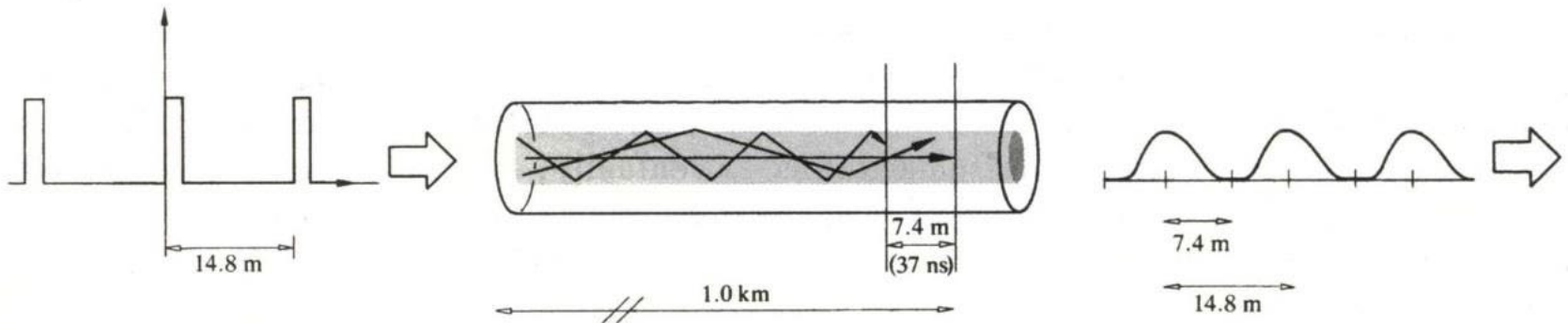
$$\Delta t = \frac{Ln_f}{c} \left(\frac{n_f}{n_c} - 1 \right)$$

The delay time between the lowest mode and highest mode:

$$\frac{\Delta t}{L} = 37 \text{ ns/km}$$

Traveling speed: $v_f = \frac{c}{n_f} = 2.0 \times 10^8 \text{ m/s}$

The spread in space: 7.4 m/km



It means that the input pulses must be at least 14.8 m apart; they must be separated in time by 74 ns and so cannot come any faster than one every 74 ns, which is a rate of 13.5 million pulses per second.

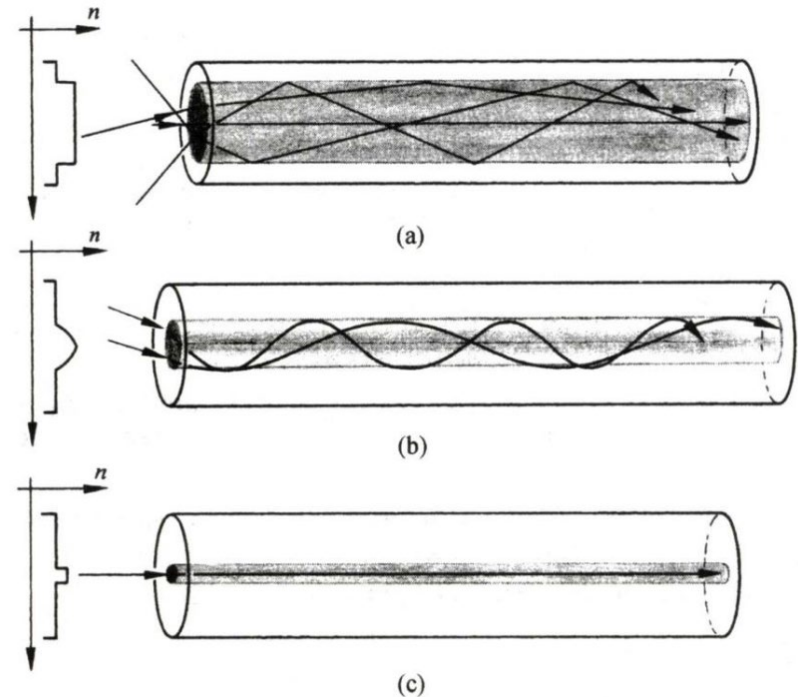
Graded-index fibers: refractive index decrease gradually from the core to the cladding.

Rays taking shorter paths are slowed down and rays spiralling around near the cladding move more swiftly over longer paths.

A graded-index fiber has a core diameter of about $20\text{ }\mu\text{m}$ to $90\text{ }\mu\text{m}$ and an intermodal dispersion of only around 2 ns/km .

Single-mode fibers: the core is so narrow (less than $10\text{ }\mu\text{m}$) that it will provide only one mode where the rays travel parallel to the central axis.

Typically having core diameters of only $2\text{ }\mu\text{m}$ to $9\text{ }\mu\text{m}$, they essentially eliminate intermodal dispersion.



- (a) Multimode step-index fiber.
- (b) Multimode graded-index fiber.
- (c) Single-mode step-index fiber.

Homework

□ 5.71, 5.73